

Fundamentals of Electrical and Electronics Engineering (FEEE)

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Lecture-20

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AC Power Analysis

1. Instantaneous Power
2. Average Power/ Effective Power / Real Power/Active Power
3. Apparent Power
4. Complex Power($\text{Real Power} + j \text{ Reactive Power}$)

1. Instantaneous Power

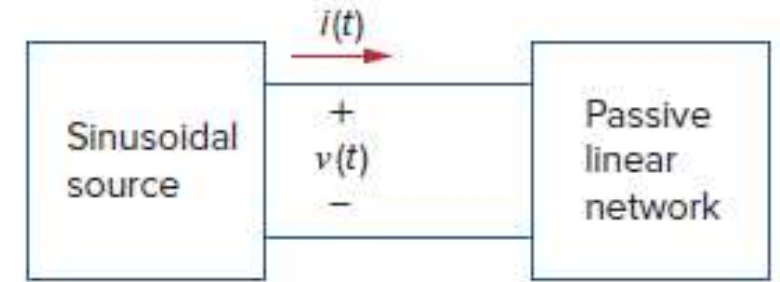
- The power absorbed by the element at a specific instant of time.
- The *instantaneous power* $p(t)$ absorbed by an element is the product of the instantaneous voltage $v(t)$ across the element and the instantaneous current $i(t)$ through it.
- The **instantaneous power** (in watts) is the power at any instant of time.

$$p(t) = v(t) i(t)$$

- It is the rate at which an element absorbs energy.

$$v(t) = V_m \cos(\omega t + \theta_v)$$

$$i(t) = I_m \cos(\omega t + \theta_i)$$



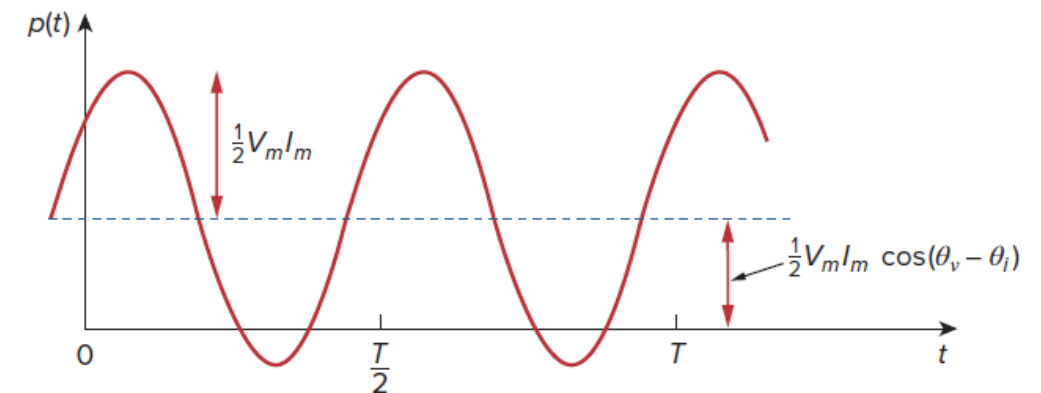
- V_m and I_m are the amplitudes (or peak values)
- θ_v and θ_i are the phase angles of the voltage and current, respectively.
- The instantaneous power absorbed by the circuit is

$$p(t) = v(t)i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

- The instantaneous power changes with time and is therefore difficult to measure. **The average power is more convenient to measure.**



2. Average Power

- The **average power**, in watts, is the average of the instantaneous power over one period.

$$P = \frac{1}{T} \int_0^T p(t) dt$$

$$P = \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) dt + \frac{1}{T} \int_0^T \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) dt$$

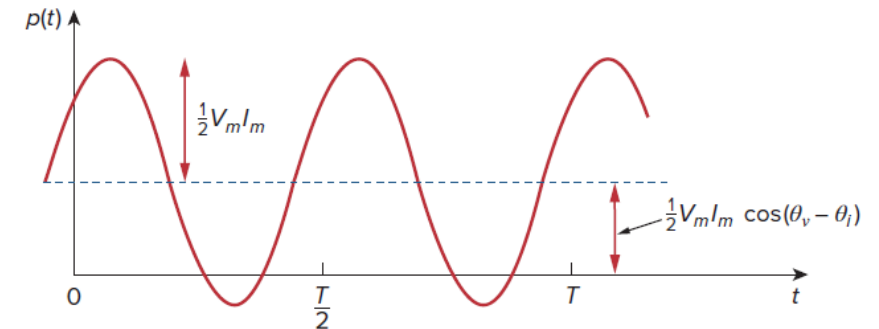
$$= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \frac{1}{T} \int_0^T dt + \frac{1}{2} V_m I_m \frac{1}{T} \int_0^T \cos(2\omega t + \theta_v + \theta_i) dt$$

Average power

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

= Effective Power / Real Power/Active Power



- The second integrand is a sinusoid. We know that the average of a sinusoid over its period is zero because the area under the sinusoid during a positive half-cycle is cancelled by the area under it during the following negative half-cycle.

average power $P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$

- Consider two special cases.
- Case 1) When $\theta_v = \theta_i$, the voltage and current are in phase. This implies a purely resistive circuit or resistive load R , and

$$P = \frac{1}{2} V_m I_m = \frac{1}{2} I_m^2 R \Rightarrow P = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} = V_{\text{rms}} I_{\text{rms}} = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R}$$

- shows that a purely resistive circuit absorbs power at all times.
- Case 2) When $\theta_v - \theta_i = \pm 90^\circ$, we have a purely reactive circuit, then

$$P = \frac{1}{2} V_m I_m \cos 90^\circ = 0$$

- showing that a purely reactive circuit absorbs no average power.
- A resistive load (R) absorbs power at all times, while a reactive load (L or C) absorbs zero average power.

1. Given that, $v(t) = 120 \cos(377t + 45^\circ)$ V and $i(t) = 10 \cos(377t - 10^\circ)$ A find the instantaneous power and the average power absorbed by the passive linear network of Fig. below.

The instantaneous power is given by

$$p = vi = 1200 \cos(377t + 45^\circ) \cos(377t - 10^\circ)$$

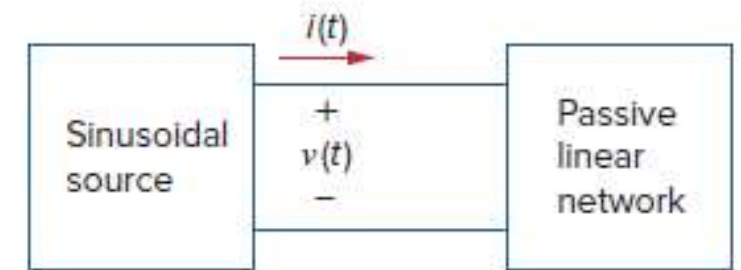
$$p = 600[\cos(754t + 35^\circ) + \cos 55^\circ]$$

The instantaneous power

$$p(t) = 344.2 + 600 \cos(754t + 35^\circ) \text{ W}$$

The average power is

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \\ &= \frac{1}{2} 120(10) \cos[45^\circ - (-10^\circ)] \\ &= 600 \cos 55^\circ = 344.2 \text{ W} \end{aligned}$$

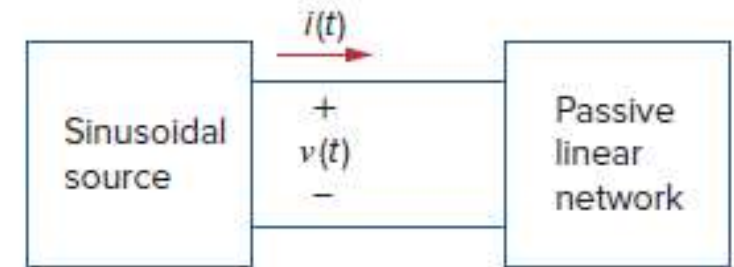


Applying the trigonometric identity

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

2. Calculate the instantaneous power and average power absorbed by the passive linear network of Fig. below. if $v(t) = 330 \cos(10t + 20^\circ)$ V and $i(t) = 33 \sin(10t + 60^\circ)$ A.



$$\text{Instantaneous power} = 3.5 + 5.445 \cos(20t - 10^\circ) \text{ kW}$$

$$\text{Average power} = 3.5 \text{ kW.}$$

3. Calculate the average power absorbed by an impedance $\mathbf{Z} = 30 - j70 \, \Omega$, when a voltage $\mathbf{V} = 120/\underline{0^\circ}$ is applied across it.

The current through the impedance is

$$\begin{aligned}\mathbf{I} &= \frac{\mathbf{V}}{\mathbf{Z}} = \frac{120/\underline{0^\circ}}{30 - j70} \\ &= \frac{120/\underline{0^\circ}}{76.16/\underline{-66.8^\circ}} \\ &= 1.576/\underline{66.8^\circ} \text{ A}\end{aligned}$$

The average power is

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} (120)(1.576) \cos(0 - 66.8^\circ) = 37.24 \text{ W}$$

4. For the circuit shown in Fig. below, find the average power supplied by the source and the average power absorbed by the resistor.

The current $\mathbf{I}$ is given by

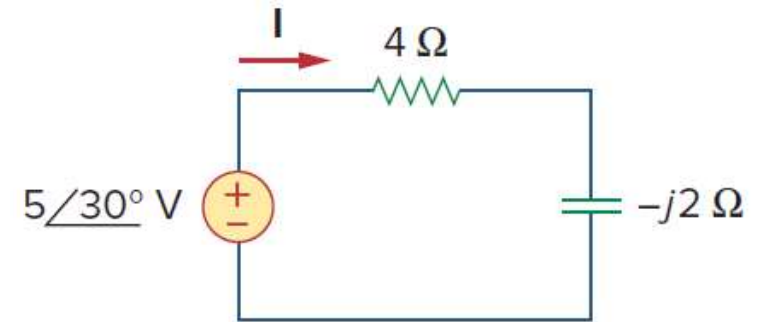
$$\mathbf{I} = \frac{5\angle 30^\circ}{4 - j2} = \frac{5\angle 30^\circ}{4.472\angle -26.57^\circ} = 1.118\angle 56.57^\circ \text{ A}$$

The average power supplied by the voltage source is

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

$$\begin{aligned} P &= \frac{1}{2} (5)(1.118) \cos(30^\circ - 56.57^\circ) \\ &= 2.5 \text{ W} \end{aligned}$$

- Average power absorbed by a resistor is the same as the average power supplied. Zero average power is absorbed by the capacitor.



The current through the resistor is

$$\mathbf{I}_R = \mathbf{I} = 1.118\angle 56.57^\circ \text{ A}$$

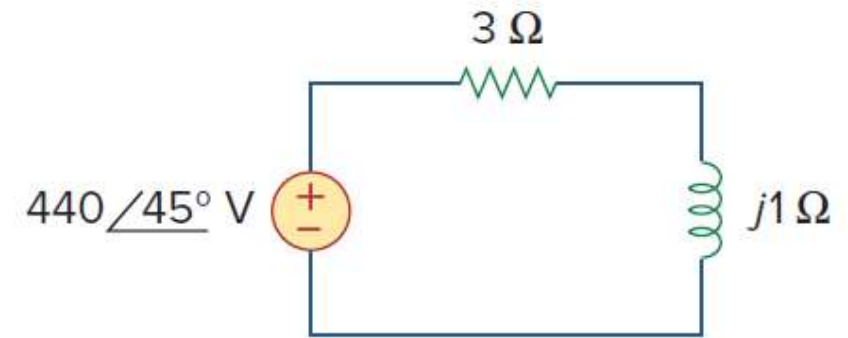
and the voltage across it is

$$\mathbf{V}_R = 4\mathbf{I}_R = 4.472\angle 56.57^\circ \text{ V}$$

The average power absorbed by the resistor is

$$P = \frac{1}{2} (4.472)(1.118) = 2.5 \text{ W}$$

5. In the circuit of Fig. below, calculate the average power absorbed by the resistor and inductor. Find the average power supplied by the voltage source.



Answer: 29.04 kW, 0 W, 29.04 kW.

6. Determine the average power generated by each source and the average power absorbed by each passive element in the circuit of Fig. below.

We apply mesh analysis as shown in Fig. (b). For mesh 1,

$$\mathbf{I}_1 = 4 \text{ A}$$

For mesh 2,

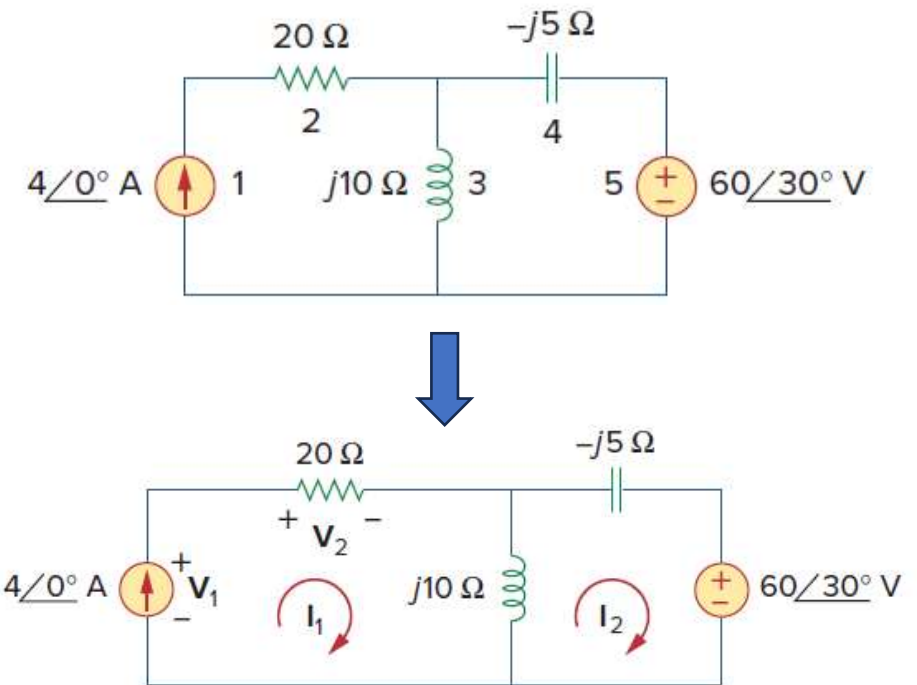
$$(j10 - j5)\mathbf{I}_2 - j10\mathbf{I}_1 + 60\angle 30^\circ = 0,$$

$$j5\mathbf{I}_2 = -60\angle 30^\circ + j40$$

$$\begin{aligned}\mathbf{I}_2 &= -12\angle -60^\circ + 8 \\ &= 10.58\angle 79.1^\circ \text{ A}\end{aligned}$$

the voltage across it is $60\angle 30^\circ \text{ V}$, so that the average power is

$$P_5 = \frac{1}{2} (60)(10.58) \cos(30^\circ - 79.1^\circ) = 207.8 \text{ W} \quad (1)\text{-Power absorbed}$$



For the resistor, the current through it is $\mathbf{I}_1 = 4\angle 0^\circ$ and the voltage across it is $20\mathbf{I}_1 = 80\angle 0^\circ$, so that the power absorbed by the resistor is

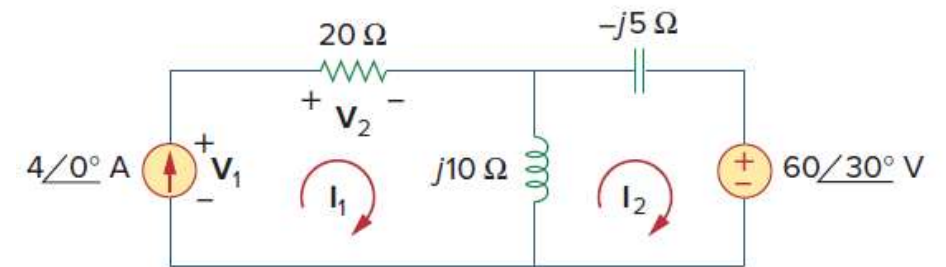
$$P_2 = \frac{1}{2} (80)(4) = 160 \text{ W} \quad (2)$$

For the current source, the current through it is $\mathbf{I}_1 = 4\angle 0^\circ$ and the voltage across it is

$$\begin{aligned} \mathbf{V}_1 &= 20\mathbf{I}_1 + j10(\mathbf{I}_1 - \mathbf{I}_2) \\ &= 80 + j10(4 - 2 - j10.39) \\ &= 183.9 + j20 = 184.984\angle 6.21^\circ \text{ V} \end{aligned}$$

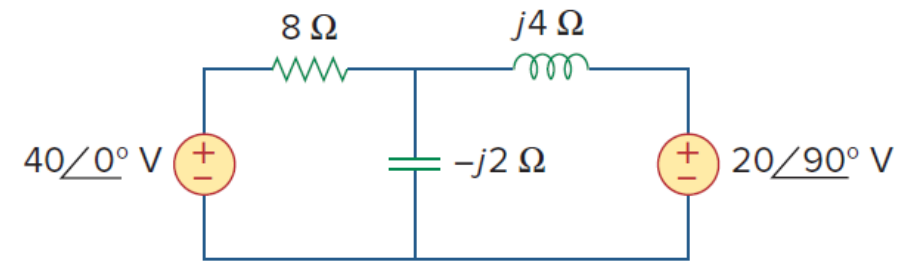
The average power supplied by the current source is

$$P_1 = \frac{1}{2} (184.984)(4) \cos(6.21^\circ - 0) = 367.8 \text{ W} \quad (3)$$



$$(1) + (2) = (3)$$

7. Calculate the average power of the five elements in the circuit of Fig. below.



Answer: 40-V Voltage source: 60 W; $j20$ -V Voltage source: 40 W; resistor: 100 W; others: 0 W.